

# Johan MAZOYER

Research Interests: Optical Instrumentation, Exoplanets, Debris Disks, Direct Imaging & Coronagraphy, Observation & Characterization of Extrasolar Systems

## 1 RESEARCH POSITIONS

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| CNRS Scientist – <b>LIRA/Paris Observatory - PSL</b> (France)            | Since 2020  |
| Carl Sagan Fellow – <b>NASA Jet Propulsion Laboratory</b> (Pasadena, CA) | 2018 - 2019 |
| Postdoc – <b>Johns Hopkins University</b> (Baltimore, MD)                | 2016 - 2018 |
| Postdoc – <b>Space Telescope Science Institute</b> (Baltimore, MD)       | 2014 - 2016 |
| Graduate Student – <b>LIRA/Paris Observatory - PSL</b> (France)          | 2011 - 2014 |

## 2 EDUCATION

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| HDR (French habilitation) – <b>Paris Observatory - PSL</b> (France)                                                                                                                            | March 2024 |
| PhD – Astronomy & Astrophysics – <b>Université Paris Cité</b> (Paris, France)<br><i>Thesis: High-Contrast Imaging Of Exoplanets And Circumstellar Disks (P. Baudoz &amp; G. Rousset)</i>       | Sept. 2014 |
| Master – Astrophysics & Planetology – <b>Université de Toulouse</b> (Toulouse, France)<br><i>Thesis: Influence of Mars atmosphere on ChemCam detection limits (O. Gasnault &amp; R. Wiens)</i> | Sept. 2011 |
| Master – Techniques for Space Imaging – <b>ISAE Supaero</b> (Toulouse, France)                                                                                                                 | Sept. 2011 |
| Bachelor – Embedded systems – <b>Ecole polytechnique</b> (Paris, France)                                                                                                                       | Sept. 2010 |

## 3 GRANTS & AWARDS

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| ERC - Consolidator Grant (PI) <b>ECHOES</b> - 2 M€                                          | Since 2026 |
| ANR / PEPR Origins (PI) <b>AMINO</b> - 1.2 M€                                               | Since 2026 |
| ANR JCJC (PI) (terminated and reimbursed) - 370 k€                                          | 2025-2026  |
| <b>Académie spatiale d'île de France</b> (PI) 1/2 PhD grant fund for L. Delaye - 60 k€/3yrs | 2025       |
| CNES (PI) ( <b>French space agency</b> ) 1/2 PhD grant for M. Castañeda Medina - 60 k€/3yrs | 2025       |
| DIM Origins (PI) Hardware funding for a spatial light modulator - 12 k€                     | 2023       |
| CNES (co-PI) ( <b>French space agency</b> ) Postdoc fund for I. Luginja - 120 k€/2 yrs      | 2022       |
| Data Intensive Artificial Intelligence program (PI) 1/2 PhD grant Y. Gutierrez - 60k€/3yrs  | 2021       |
| <b>Ecosud</b> Program (PI) Collaboration <i>Universidad de Chile</i> – 50 k€                | 2020       |
| NASA Group Award: LBTI Hosts Survey Science Team                                            | 2020       |
| Carl Sagan Fellowship ( <b>NASA Hubble Fellowship Program</b> ) – 280k€/3 yrs               | 2018       |
| Cover of <b>Astronomy &amp; Astrophysics</b> Journal ( <b>Volume 564</b> )                  | 2014       |
| Thesis Award (CNES fellow symposium JC2)                                                    | 2013       |

## 4 OUTREACH



I am heavily involved in scientific outreach in France. In addition to regular class or public interventions, I am running institutional events and associative initiatives:

- **Podcast Science:** I am running **PodcastScience.fm**, a **general science program**, airing every weeks, in french. Listened by 10'000 to 20'000 listeners, Podcast Science received the Golden blog award for best scientific blog in 2012.

- **Les p'tits cueilleurs d'étoiles:** Association organizing astronomers visits in children hospitals in France. I am organizing all Paris region hospitals (25 visits/yr).

- **Fête de la science:** I was the main organizer of the annual Paris Observatory Doors Open Days (~1000 visitors each year) for two consecutive years (2023 and 2024)

## 5 PROFESSIONAL ACTIVITIES & SERVICE

### Responsibilities in scientific instruments:

- **Roman Space Telescope Coronagraph:** Deputy CNES representative Since 2023
- **VLT/SPHERE+:** Dark-Hole work package leader Since 2022
- **Habitable Exoplanet Observatory (HabEx):** Contributing Scientist 2019
- **Large UV Optical Infrared Surveyor (LUVOIR):** Contributing Scientist 2019
- **Gemini Planet Imager (GPI):** junior member of the consortium 2017-2020

### Organization of Scientific Meetings:

- HWO 2026 Annual Symposium (LOC) Massy, 2026
- Roman coronagraphic instrument summer school (SOC) Nice, March 2026
- ExoSystèmes 4 (SOC) Lyon, 2024
- National Capital Area Disks conference (SOC & LOC) Baltimore 2018
- Optimal Optical Coronagraphs workshop (SOC & LOC) Leiden, 2017
- High Contrast Imaging from Space (SOC) Baltimore, 2016

### Other services:

- **"Exoplanetary systems" Group Leader** at Paris Observatory / LIRA Since 2025
- **Hubble Telescope Allocation Committee** 2024
- Science Commity of CNRS/INSU's exoplanets group (**CET exoplanètes**) 2023 - 2024
- Science Commity of CNRS/INSU's High Angular Resolution Group (**CS-ASHRA**) Since 2021
- **Reviewer** for *AJ*, *A&A*, *MNRAS*, *PASP* and *JATIS*.
- **External Reviewer** for 2 Phd Committees (France, Belgium).

## 6 MENTORING

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|-------------------------------------------------------------------------------------------|------------|
| Manuela Castañeda Medina (PhD, CNES/ONERA): co-direction with L. Mugnier                  | Since 2026 |
| Lukas Delaye (PhD, LIRA): co-director with A. Potier                                      | Since 2025 |
| Yann Gutierrez (PhD, LIRA): co-director with L. Mugnier, ONERA                            | 2022-2025  |
| Vito Squicciarini (Postdoc, LIRA): co-advisor with A.-M. Lagrange                         | 2022-2024  |
| Iva Leginja (Postdoc, LIRA): main advisor ; CNES Fellow. Now Resercher at Obs Côte D'Azur | 2022-2024  |
| Sophia Stasevic (PhD, LIRA) co-director with A.-M. Lagrange and J. Milli                  | 2021-2025  |
| Justin Hom (PhD, ASU) co-advisor with J. Patience                                         | 2019-2023  |
| Kevin Fogarty (PhD, JHU) co-advisor with L. Pueyo. Now Research Scientist at AMES         | 2017-2019  |

## 7 TEACHING

### Observatoire de Paris Master Class:

- Instrumentation for Astronomy
- Detection of Exoplanets

## RESEARCH ACHIEVEMENTS

I have a longstanding interest in developing innovative high-contrast techniques, which are critical for detecting and characterizing faint objects orbiting closely to bright stars (exoplanets and dust belts). Over the years, I have gained extensive expertise in the analytical understanding and numerical simulation of these systems for both space- and ground-based instruments, achieving unprecedented performance on experimental testbeds and on-sky. Additionally, I am deeply engaged in exosystem science, having analyzed coronagraphic data from various high-contrast instruments. My research complementary integrates instrumental development and observational studies. As coronagraphic instruments grow in complexity, maximizing their potential requires instrumentation expertise, and the insights gained from on-sky observations are essential for shaping future instrumental advancements.

**62 accepted papers in peer-reviewed journals, including:**

- **7 papers as first author + 1 as co-first author,**
- **10 papers with major contributions (2nd or 3rd author),**
- **44 papers with more minor contributions.**

**49 conference proceedings (mainly SPIE), including 10 as first author.**

**All my papers and proceedings are in open access.**

**Google Scholar analytics: 4012 citations – h-index = 35.**

## MAJOR REFEREED PUBLICATIONS

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1. Squicciarini, V. ; **Mazoyer, J.** ; Wilkinson, C. et al. (2025), *GPI+SPHERE detection of a 6.1  $M_{Jup}$  circumbinary planet around HD 143811*, Astronomy and Astrophysics, 702, L10, [DOI link](#), [arXiv link](#), 7 citations
2. Stasevic, S. ; Milli, J. ; **Mazoyer, J.** et al. (2025), *Optimising reference library selection for reference-star differential imaging of discs with SPHERE/IRDIS*, Astronomy and Astrophysics, 701, A93, [DOI link](#), [arXiv link](#)
3. Leginja, I. ; Baudoz, P. ; **Mazoyer, J.** et al. (2025), *Extended linearity in the high-order wavefront sensor for the Roman Coronagraph*, Astronomy and Astrophysics, 698, A130, [DOI link](#), [arXiv link](#), 5 citations
4. Squicciarini, V. ; **Mazoyer, J.** ; Lagrange, A. -M. et al. (2025), *The COBREX archival survey: Improved constraints on the occurrence rate of wide-orbit substellar companions: I. A uniform re-analysis of 400 stars from the GPIES survey*, Astronomy and Astrophysics, 693, A54, [DOI link](#), [arXiv link](#), 13 citations
5. Gutierrez, Y. ; **Mazoyer, J.** ; Mugnier, L. M. et al. (2024), *Image-based wavefront correction using model-free reinforcement learning*, Optics Express, 32, 31247, [DOI link](#), [arXiv link](#), 2 citations
6. Galicher, R. ; Potier, A. ; **Mazoyer, J.** et al. (2024), *Increasing the raw contrast of VLT/SPHERE with the dark hole technique. III. Broadband reference differential imaging of HR4796 using a four-quadrant phase mask*, Astronomy and Astrophysics, 686, A54, [DOI link](#), [arXiv link](#), 6 citations
7. Galicher, R. & **Mazoyer, J.** (2024), *Imaging exoplanets with coronagraphic instruments*, Comptes Rendus Physique, 24, 133, [DOI link](#), [arXiv link](#), 23 citations
8. Stasevic, S. ; Milli, J. ; **Mazoyer, J.** et al. (2023), *An inner warp discovered in the disk around HD 110058 using VLT/SPHERE and HST/STIS*, Astronomy and Astrophysics, 678, A8, [DOI link](#), [arXiv link](#), 10 citations
9. Potier, A. ; **Mazoyer, J.** ; Wahhaj, Z. et al. (2022), *Increasing the raw contrast of VLT/SPHERE with the dark hole technique. II. On-sky wavefront correction and coherent differential imaging*, Astronomy and Astrophysics, 665, A136, [DOI link](#), [arXiv link](#), 29 citations
10. Chen, C. ; **Mazoyer, J.** ; Poteet, C. A. et al. (2020), *Multiband GPI Imaging of the HR 4796A Debris Disk*, The Astrophysical Journal, 898, 55, [DOI link](#), [arXiv link](#), 42 citations

11. **Mazoyer, J.** ; Pueyo, L. ; N'Diaye, M. et al. (2018), *Active Correction of Aperture Discontinuities-Optimized Stroke Minimization. II. Optimization for Future Missions*, The Astronomical Journal, 155, 8, [DOI link](#), [arXiv link](#), 24 citations
12. **Mazoyer, J.** ; Pueyo, L. ; N'Diaye, M. et al. (2018), *Active Correction of Aperture Discontinuities-Optimized Stroke Minimization. I. A New Adaptive Interaction Matrix Algorithm*, The Astronomical Journal, 155, 7, [DOI link](#), [arXiv link](#), 18 citations
13. Fogarty, K. ; Pueyo, L. ; **Mazoyer, J.** et al. (2017), *Polynomial Apodizers for Centrally Obscured Vortex Coronagraphs*, The Astronomical Journal, 154, 240, [DOI link](#), [arXiv link](#), 10 citations
14. **Mazoyer, J.** ; Pueyo, L. ; Norman, C. et al. (2016), *Active compensation of aperture discontinuities for WFIRST-AFTA: analytical and numerical comparison of propagation methods and preliminary results with a WFIRST-AFTA-like pupil*, Journal of Astronomical Telescopes, Instruments, and Systems, 2, 011008, [DOI link](#), [arXiv link](#), 9 citations
15. **Mazoyer, J.** ; Boccaletti, A. ; Choquet, É. et al. (2016), *A Symmetric Inner Cavity in the HD 141569A Circumstellar Disk*, The Astrophysical Journal, 818, 150, [DOI link](#), [arXiv link](#), 14 citations
16. **Mazoyer, J.** ; Boccaletti, A. ; Augereau, J. -C. et al. (2014), *Is the HD 15115 inner disk really asymmetrical?*, Astronomy and Astrophysics, 569, A29, [DOI link](#), [arXiv link](#), 35 citations
17. **Mazoyer, J.** ; Baudoz, P. ; Galicher, R. et al. (2014), *High-contrast imaging in polychromatic light with the self-coherent camera*, Astronomy and Astrophysics, 564, L1, [DOI link](#), [arXiv link](#), 37 citations
18. **Mazoyer, J.** ; Baudoz, P. ; Galicher, R. et al. (2013), *Estimation and correction of wavefront aberrations using the self-coherent camera: laboratory results*, Astronomy and Astrophysics, 557, A9, [DOI link](#), [arXiv link](#), 41 citations

## OTHER REFEREED PUBLICATIONS

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1. Landman, R. ; Doelman, D. ; Rietjens, J. et al. (2026), *SUPPPRESS: Prototyping and testing liquid-crystal vector vortex coronagraphs with reduced polarization leakage*, accepted in JATIS, [arXiv link](#)
2. Goulas, C. ; Galicher, R. ; Vidal, F. et al. (2026), *Non-common path aberration compensation and a dark hole loop with a pyramid adaptive optics system: Application to SAXO+*, Astronomy and Astrophysics, 708, A50, [DOI link](#), [arXiv link](#)
3. Potier, A. ; Galicher, R. ; Baudoz, P. et al. (2025), *Coherent differential imaging of high-contrast extended sources with VLT/SPHERE*, Astronomy and Astrophysics, 704, A98, [DOI link](#), [arXiv link](#)
4. Hom, J. ; Esposito, T. M. ; Crotts, K. A. et al. (2025), *The Disks In Scorpius-Centaurus Survey (DISCS). I. Four Newly Resolved Debris Disks in Polarized Intensity Light*, The Astronomical Journal, 170, 46, [DOI link](#), [arXiv link](#), 2 citations
5. Lagrange, A. -M. ; Wilkinson, C. ; Mâlin, M. et al. (2025), *Evidence for a sub-Jovian planet in the young TWA 7 disk*, Nature, 642, 905, [DOI link](#), [arXiv link](#), 39 citations
6. Desgrange, C. ; Milli, J. ; Chauvin, G. et al. (2025), *Dust populations from 30 to 1000 au in the debris disk of HD 120326: Panchromatic view with VLT/SPHERE, ALMA, and HST/STIS*, Astronomy and Astrophysics, 698, A183, [DOI link](#), [arXiv link](#), 5 citations
7. Chomez, A. ; Delorme, P. ; Lagrange, A. -M. et al. (2025), *The SPHERE infrared survey for exoplanets (SHINE): IV. Complete observations, data reduction and analysis, detection performances, and final results*, Astronomy and Astrophysics, 697, A99, [DOI link](#), [arXiv link](#), 12 citations
8. Ray, S. ; Sallum, S. ; Hinkley, S. et al. (2025), *The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems. III. Aperture Masking Interferometric Observations of the Star HIP 65426 at 3.8  $\mu\text{m}$* , The Astrophysical Journal Letters, 983, L25, [DOI link](#), [arXiv link](#), 11 citations
9. Lajinja, I. ; Carrión-González, Ó. ; Laugier, R. et al. (2025), *Advancing European high-contrast imaging R&D towards the Habitable Worlds Observatory*, Astrophysics and Space Science, 370, 29, [DOI link](#), [arXiv link](#), 2 citations

10. Wilkinson, C. ; Charnay, B. ; Mazevet, S. et al. (2024), *Breaking degeneracies in exoplanetary parameters through self-consistent atmosphere-interior modelling*, *Astronomy and Astrophysics*, 692, A113, [DOI link](#), [arXiv link](#), 14 citations
11. Lewis, B. L. ; Fitzgerald, M. P. ; Esposito, T. M. et al. (2024), *Gemini Planet Imager Observations of a Resolved Low-inclination Debris Disk around HD 156623*, *The Astronomical Journal*, 168, 142, [DOI link](#), [arXiv link](#), 2 citations
12. Goulas, C. ; Galicher, R. ; Vidal, F. et al. (2024), *Numerical simulations for the SAXO+ upgrade: Performance analysis of the adaptive optics system*, *Astronomy and Astrophysics*, 689, A199, [DOI link](#), [arXiv link](#), 5 citations
13. Petrus, S. ; Whiteford, N. ; Patapis, P. et al. (2024), *The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems. V. Do Self-consistent Atmospheric Models Represent JWST Spectra? A Showcase with VHS 1256-1257 b*, *The Astrophysical Journal Letters*, 966, L11, [DOI link](#), [arXiv link](#), 42 citations
14. Hom, J. ; Patience, J. ; Chen, C. H. et al. (2024), *A uniform analysis of debris discs with the Gemini Planet Imager II: constraints on dust density distribution using empirically informed scattering phase functions*, *Monthly Notices of the Royal Astronomical Society*, 528, 6959, [DOI link](#), [arXiv link](#), 12 citations
15. Sallum, S. ; Ray, S. ; Kammerer, J. et al. (2024), *The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems. IV. NIRISS Aperture Masking Interferometry Performance and Lessons Learned*, *The Astrophysical Journal Letters*, 963, L2, [DOI link](#), [arXiv link](#), 15 citations
16. Worthen, K. ; Chen, C. H. ; Brittain, S. D. et al. (2024), *Vertical Structure of Gas and Dust in Four Debris Disks*, *The Astrophysical Journal*, 962, 166, [DOI link](#), [arXiv link](#), 6 citations
17. Crotts, K. A. ; Matthews, B. C. ; Duchêne, G. et al. (2024), *A Uniform Analysis of Debris Disks with the Gemini Planet Imager. I. An Empirical Search for Perturbations from Planetary Companions in Polarized Light Images*, *The Astrophysical Journal*, 961, 245, [DOI link](#), [arXiv link](#), 19 citations
18. Vaughan, S. R. ; Gebhard, T. D. ; Bott, K. et al. (2023), *Chasing rainbows and ocean glints: Inner working angle constraints for the Habitable Worlds Observatory*, *Monthly Notices of the Royal Astronomical Society*, 524, 5477, [DOI link](#), [arXiv link](#), 54 citations
19. Carter, A. L. ; Hinkley, S. ; Kammerer, J. et al. (2023), *The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems I: High-contrast Imaging of the Exoplanet HIP 65426 b from 2 to 16  $\mu\text{m}$* , *The Astrophysical Journal Letters*, 951, L20, [DOI link](#), [arXiv link](#), 116 citations
20. Miles, B. E. ; Biller, B. A. ; Patapis, P. et al. (2023), *The JWST Early-release Science Program for Direct Observations of Exoplanetary Systems II: A 1 to 20  $\mu\text{m}$  Spectrum of the Planetary-mass Companion VHS 1256-1257 b*, *The Astrophysical Journal Letters*, 946, L6, [DOI link](#), [arXiv link](#), 182 citations
21. Hinkley, S. ; Carter, A. L. ; Ray, S. et al. (2022), *The JWST Early Release Science Program for the Direct Imaging and Spectroscopy of Exoplanetary Systems*, *Publications of the Astronomical Society of the Pacific*, 134, 095003, [DOI link](#), [arXiv link](#), 57 citations
22. Crotts, K. A. ; Draper, Z. H. ; Matthews, B. C. et al. (2022), *A Multiwavelength Study of the Highly Asymmetrical Debris Disk around HD 111520*, *The Astrophysical Journal*, 932, 23, [DOI link](#), [arXiv link](#), 9 citations
23. Betti, S. K. ; Follette, K. ; Jorquera, S. et al. (2022), *Detection of Near-infrared Water Ice at the Surface of the (Pre)Transitional Disk of AB Aur: Informing Icy Grain Abundance, Composition, and Size*, *The Astronomical Journal*, 163, 145, [DOI link](#), [arXiv link](#), 20 citations
24. Singh, G. ; Bhowmik, T. ; Boccaletti, A. et al. (2021), *Revealing asymmetrical dust distribution in the inner regions of HD 141569*, *Astronomy and Astrophysics*, 653, A79, [DOI link](#), [arXiv link](#), 15 citations
25. Crotts, K. A. ; Matthews, B. C. ; Esposito, T. M. et al. (2021), *A Deep Polarimetric Study of the Asymmetrical Debris Disk HD 106906*, *The Astrophysical Journal*, 915, 58, [DOI link](#), [arXiv link](#), 16 citations

26. Arriaga, P. ; Fitzgerald, M. P. ; Duchêne, G. et al. (2020), *Multiband Polarimetric Imaging of HR 4796A with the Gemini Planet Imager*, The Astronomical Journal, 160, 79, [DOI link](#), [arXiv link](#), 36 citations
27. Esposito, T. M. ; Kalas, P. ; Fitzgerald, M. P. et al. (2020), *Debris Disk Results from the Gemini Planet Imager Exoplanet Survey's Polarimetric Imaging Campaign*, The Astronomical Journal, 160, 24, [DOI link](#), [arXiv link](#), 112 citations
28. Duchêne, G. ; Rice, M. ; Hom, J. et al. (2020), *The Gemini Planet Imager View of the HD 32297 Debris Disk*, The Astronomical Journal, 159, 251, [DOI link](#), [arXiv link](#), 27 citations
29. Ertel, S. ; Defrère, D. ; Hinz, P. et al. (2020), *The HOSTS Survey for Exozodiacal Dust: Observational Results from the Complete Survey*, The Astronomical Journal, 159, 177, [DOI link](#), [arXiv link](#), 135 citations
30. Bruzzone, J. S. ; Metchev, S. ; Duchêne, G. et al. (2020), *Imaging the 44 au Kuiper Belt Analog Debris Ring around HD 141569A with GPI Polarimetry*, The Astronomical Journal, 159, 53, [DOI link](#), [arXiv link](#), 12 citations
31. Hom, J. ; Patience, J. ; Esposito, T. M. et al. (2020), *First Resolved Scattered-light Images of Four Debris Disks in Scorpius-Centaurus with the Gemini Planet Imager*, The Astronomical Journal, 159, 31, [DOI link](#), [arXiv link](#), 16 citations
32. Bhowmik, T. ; Boccaletti, A. ; Thébault, P. et al. (2019), *Spatially resolved spectroscopy of the debris disk HD 32297. Further evidence of small dust grains*, Astronomy and Astrophysics, 630, A85, [DOI link](#), [arXiv link](#), 38 citations
33. Ren, B. ; Choquet, É. ; Perrin, M. D. et al. (2019), *An Exo-Kuiper Belt with an Extended Halo around HD 191089 in Scattered Light*, The Astrophysical Journal, 882, 64, [DOI link](#), [arXiv link](#), 44 citations
34. Stark, C. C. ; Belikov, R. ; Bolcar, M. R. et al. (2019), *ExoEarth yield landscape for future direct imaging space telescopes*, Journal of Astronomical Telescopes, Instruments, and Systems, 5, 024009, [DOI link](#), [arXiv link](#), 87 citations
35. Engler, N. ; Boccaletti, A. ; Schmid, H. M. et al. (2019), *Investigating the presence of two belts in the HD 151115 system*, Astronomy and Astrophysics, 622, A192, [DOI link](#), [arXiv link](#), 35 citations
36. Esposito, T. M. ; Duchêne, G. ; Kalas, P. et al. (2018), *Direct Imaging of the HD 35841 Debris Disk: A Polarized Dust Ring from Gemini Planet Imager and an Outer Halo from HST/STIS*, The Astronomical Journal, 156, 47, [DOI link](#), [arXiv link](#), 35 citations
37. Lebouilleux, L. ; Sauvage, J. -F. ; Pueyo, L. A. et al. (2018), *Pair-based Analytical model for Segmented Telescopes Imaging from Space for sensitivity analysis*, Journal of Astronomical Telescopes, Instruments, and Systems, 4, 035002, [DOI link](#), [arXiv link](#), 21 citations
38. Poteet, C. A. ; Chen, C. H. ; Hines, D. C. et al. (2018), *Space-based Coronagraphic Imaging Polarimetry of the TW Hydrae Disk: Shedding New Light on Self-shadowing Effects*, The Astrophysical Journal, 860, 115, [DOI link](#), [arXiv link](#), 13 citations
39. Jensen-Clem, R. ; Mawet, D. ; Gomez Gonzalez, C. A. et al. (2018), *A New Standard for Assessing the Performance of High Contrast Imaging Systems*, The Astronomical Journal, 155, 19, [DOI link](#), [arXiv link](#), 34 citations
40. Perrot, C. ; Boccaletti, A. ; Pantin, E. et al. (2016), *Discovery of concentric broken rings at sub-arcsec separations in the HD 141569A gas-rich, debris disk with VLT/SPHERE*, Astronomy and Astrophysics, 590, L7, [DOI link](#), [arXiv link](#), 48 citations
41. Delorme, J. R. ; Galicher, R. ; Baudoz, P. et al. (2016), *Focal plane wavefront sensor achromatization: The multireference self-coherent camera*, Astronomy and Astrophysics, 588, A136, [DOI link](#), [arXiv link](#), 19 citations
42. Choquet, É. ; Perrin, M. D. ; Chen, C. H. et al. (2016), *First Images of Debris Disks around TWA 7, TWA 25, HD 35650, and HD 377*, The Astrophysical Journal Letters, 817, L2, [DOI link](#), [arXiv link](#), 88 citations
43. Debes, J. H. ; Ygouf, M. ; Choquet, E. et al. (2016), *Wide-Field Infrared Survey Telescope-Astrophysics Focused Telescope Assets coronagraphic operations: lessons learned from the Hubble Space Telescope*

- and the James Webb Space Telescope, *Journal of Astronomical Telescopes, Instruments, and Systems*, 2, 011010, [DOI link](#), [arXiv link](#), 12 citations
44. Wiens, R. C. ; Maurice, S. ; Lasue, J. et al. (2013), *Pre-flight calibration and initial data processing for the ChemCam laser-induced breakdown spectroscopy instrument on the Mars Science Laboratory rover*, *Spectrochimica Acta - Part B: Atomic Spectroscopy*, 82, 1, [DOI link](#), [HAL notice](#), 184 citations
  45. Cousin, A. ; Forni, O. ; Maurice, S. et al. (2011), *Laser induced breakdown spectroscopy library for the Martian environment*, *Spectrochimica Acta - Part B: Atomic Spectroscopy*, 66, 805, [DOI link](#), [HAL notice](#), 63 citations

## MAJOR CONFERENCE PROCEEDINGS

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1. Mazoyer, J. ; Mellado-Tenorio, M. J. ; Potier, A. et al. (2026), *Coherence differential imaging using gradient-boosted decision trees for the direct detection of exoplanets*, *Space Telescopes and Instrumentation 2026: Optical, Infrared, and Millimeter Wave*, [DOI link](#), [arXiv link](#)
2. Mazoyer, J. ; Goulas, C. ; Galicher, R. et al. (2026), *SAXO+, the second-stage adaptive optics for SPHERE: NCPA compensation and dark-hole loop with a pyramid wavefront sensor*, *Adaptive Optics Systems X*, [DOI link](#), [arXiv link](#)
3. Gutierrez, Y. ; Mazoyer, J. ; Herscovici-Schiller, O. et al. (2024), *A deep reinforcement learning approach to wavefront control for exoplanet imaging*, *Space Telescopes and Instrumentation 2024: Optical, Infrared, and Millimeter Wave*, 13092, 130926H, [DOI link](#), [arXiv link](#), 2 citations
4. Mazoyer, J. ; Goulas, C. ; Vidal, F. et al. (2024), *Upgrading SPHERE with the second stage AO system SAXO+: non-common path aberrations estimation and correction*, *Ground-based and Airborne Instrumentation for Astronomy X*, 13096, 130969D, [DOI link](#), [arXiv link](#), 2 citations
5. Fogarty, K. ; Mawet, D. ; Mazoyer, J. et al. (2020), *Towards high throughput and low-order aberration robustness for vortex coronagraphs with central obstructions*, *Space Telescopes and Instrumentation 2020: Optical, Infrared, and Millimeter Wave*, 11443, 114433Y, [DOI link](#), [HAL notice](#), 1 citation
6. Mazoyer, J. ; Arriaga, P. ; Hom, J. et al. (2020), *DiskFM: A forward modeling tool for disk analysis with coronagraphic instruments*, *Ground-based and Airborne Instrumentation for Astronomy VIII*, 11447, 1144759, [DOI link](#), [arXiv link](#), 12 citations
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8. Ruane, G. ; Riggs, A. ; Mazoyer, J. et al. (2018), *Review of high-contrast imaging systems for current and future ground- and space-based telescopes I: coronagraph design methods and optical performance metrics*, *Space Telescopes and Instrumentation 2018: Optical, Infrared, and Millimeter Wave*, 10698, 106982S, [DOI link](#), [arXiv link](#), 22 citations
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